Supplementary Data Files For The Expanded Public-Data Fetal Growth Analysis

Morsal Mubarak

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1 Supplementary Tables

Table 1: Expanded public data inventory used in the public-data validation framework.

Source	Files/years	Raw rows	Analysis rows	Role
NCHS cohort infant death numerator	8	330851	143079	record-level aggregate analysis
NCHS cohort linked denominator	8	29950846	28872085	record-level aggregate analysis
NCHS fetal death	11	498185	222075	record-level aggregate analysis
NCHS linked denominator	8	29633197	28644762	record-level aggregate analysis
NCHS linked infant death numerator	8	163804	141354	record-level aggregate analysis
NCHS natality	9	33589282	32448118	record-level aggregate analysis
who	1	1944	1944	external fetal-growth standards comparison
intergrowth_biometry	1	945	945	external fetal-growth standards comparison
intergrowth_efw	1	168	168	external fetal-growth standards comparison
nichd	1	25200	25200	external fetal-growth standards comparison
CDC WONDER natality expanded	1	10658	10658	stage-1 aggregate cross-check and reproducibility bridge
CDC WONDER linked birth-infant death expanded	1	3106	3106	stage-1 aggregate cross-check and reproducibility bridge
CDC WONDER fetal deaths expanded	1	5807	5807	stage-1 aggregate cross-check and reproducibility bridge
Michaelson/Grantz/research PDFs	10	339	339	conceptual framework and literature basis

Table 2: CDC WONDER aggregate exports retained as stage-1 cross-checks.

Source	Rows	Count field	Count sum	Role
CDC WONDER natality expanded	10658	Births	32393544	stage-1 aggregate cross-check and reproducibility bridge
CDC WONDER linked birth-infant death expanded	3106	Deaths	110778	stage-1 aggregate cross-check and reproducibility bridge
CDC WONDER fetal deaths expanded	5807	Fetal Deaths	194559	stage-1 aggregate cross-check and reproducibility bridge

Table 3: Record counts by parsed public-use file.

Source	Year	ZIP file	Records	Singleton records	Analysis records	resident records	period year
NCHS cohort infant death numerator	2016	2017PE2016CO.zip	23079	17122	16928	20157.0	2017.0
NCHS cohort infant death numerator	2016	2017PE2016CO.zip	2280	2519	2504	2736.0	2017.0
NCHS cohort infant death numerator	2017	2018PE2017CO.zip	22280	16582	16389	19513.0	2018.0
NCHS cohort infant death numerator	2017	2018PE2017CO.zip	2390	2444	2433	2654.0	2018.0
NCHS cohort infant death numerator	2018	2019PE2018CO.zip	21390	16065	15898	18703.0	2019.0
NCHS cohort infant death numerator	2018	2019PE2018CO.zip	2051	2445	2440	2643.0	2019.0
NCHS cohort infant death numerator	2019	2020PE2019CO.zip	20514	15526	15374	18129.0	2020.0
NCHS cohort infant death numerator	2019	2020PE2019CO.zip	1947	2321	2315	2510.0	2020.0
NCHS cohort infant death numerator	2020	2021PE2020CO.zip	19475	14577	14417	16936.0	2021.0
NCHS cohort infant death numerator	2020	2021PE2020CO.zip	1772	2209	2207	2410.0	2021.0
NCHS cohort infant death numerator	2021	2022PE2021CO.zip	17728	15044	14877	17302.0	2022.0
NCHS cohort infant death numerator	2021	2022PE2021CO.zip	2033	2452	2448	2663.0	2022.0
NCHS cohort infant death numerator	2022	2023PE2022CO.zip	20334	15400	15233	17640.0	2023.0
NCHS cohort infant death numerator	2022	2023PE2022CO.zip	1947	2440	2433	2628.0	2023.0

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Table 3: Record counts by parsed public-use file.

Source	Year	ZIP file	Records	Singleton records	Analysis records	resident records	period year
NCHS cohort infant death numerator	2023	2024PE2023CO.zip	19947	15078	14910	17291.0	2024.0
NCHS cohort infant death numerator	2023	2024PE2023CO.zip	2836	2277	2273	2452.0	2024.0
NCHS cohort linked denominator	2016	2017PE2016CO.zip	3956112	3810149	3804446	3945875.0	2017.0
NCHS cohort linked denominator	2017	2018PE2017CO.zip	3864754	3723273	3719453	3855500.0	2018.0
NCHS cohort linked denominator	2018	2019PE2018CO.zip	3801534	3664651	3661010	3791712.0	2019.0
NCHS cohort linked denominator	2019	2020PE2019CO.zip	3757582	3623963	3619959	3747540.0	2020.0
NCHS cohort linked denominator	2020	2021PE2020CO.zip	3619826	3498335	3494952	3613647.0	2021.0
NCHS cohort linked denominator	2021	2022PE2021CO.zip	3669928	3547198	3544282	3664292.0	2022.0
NCHS cohort linked denominator	2022	2023PE2022CO.zip	3676029	3550380	3547605	3667758.0	2023.0
NCHS cohort linked denominator	2023	2024PE2023CO.zip	3605081	3482971	3480378	3596017.0	2024.0
NCHS fetal death	2014	Fetal2014US_CO.zip	52372	46122	21473		
NCHS fetal death	2015	Fetal2015US_CO.zip	51490	48275	21575		
NCHS fetal death	2016	Fetal2016US_CO.zip	51391	47811	21779		
NCHS fetal death	2017	Fetal2017US_CO.zip	49170	44003	21074		
NCHS fetal death	2018	Fetal2018US_CO.zip	47876	42745	20868		
NCHS fetal death	2019	Fetal2019US_CO.zip	46007	41496	20015		
NCHS fetal death	2020	Fetal2020US_CO.zip	43146	38341	19466		
NCHS fetal death	2021	Fetal2021US_CO.zip	44428	39133	19672		
NCHS fetal death	2022	Fetal2022US_CO.zip	40113	36789	18924		
NCHS fetal death	2023	Fetal2023US_CO.zip	39574	36325	18577		
NCHS fetal death	2024	Fetal2024US_CO.zip	35648	32747	18652		
NCHS linked denominator	2017	2017PE2016CO.zip	3864781	3732243	3728351		
NCHS linked denominator	2018	2018PE2017CO.zip	3801534	3674115	3670407		
NCHS linked denominator	2019	2019PE2018CO.zip	3757582	3633587	3629448		
NCHS linked denominator	2020	2020PE2019CO.zip	3619826	3504248	3500729		

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Table 3: Record counts by parsed public-use file.

Source	Year	ZIP file	Records	Singleton records	Analysis records	resident records	period year
NCHS linked denominator	2021	2021PE2020CO	3669928	3552604	3549554		
NCHS linked denominator	2022	2022PE2021CO	3676029	3558362	3555454		
NCHS linked denominator	2023	2023PE2022CO	3605081	3491735	3488996		
NCHS linked denominator	2024	2024PE2023CO	3638436	3526370	3521823		
NCHS linked infant death numerator	2017	2017PE2016CO	22480	19129	18913		
NCHS linked infant death numerator	2018	2018PE2017CO	21890	18534	18356		
NCHS linked infant death numerator	2019	2019PE2018CO	21014	18002	17844		
NCHS linked infant death numerator	2020	2020PE2019CO	19475	16921	16753		
NCHS linked infant death numerator	2021	2021PE2020CO	19728	17268	17098		
NCHS linked infant death numerator	2022	2022PE2021CO	20334	17881	17708		
NCHS linked infant death numerator	2023	2023PE2022CO	19947	17542	17367		
NCHS linked infant death numerator	2024	2024PE2023CO	19836	17524	17315		
NCHS natality	2016	Nat2016us.zip	3956112	3820036	3813552		
NCHS natality	2017	Nat2017us.zip	3864754	3732218	3727649		
NCHS natality	2018	Nat2018us.zip	3801534	3674115	3669736		
NCHS natality	2019	Nat2019us.zip	3757582	3633587	3628705		
NCHS natality	2020	Nat2020us.zip	3619826	3504248	3499833		
NCHS natality	2021	Nat2021us.zip	3669928	3552604	3547665		
NCHS natality	2022	Nat2022us.zip	3676029	3558362	3553730		
NCHS natality	2023	Nat2023us.zip	3605081	3491735	3487259		
NCHS natality	2024	Nat2024us.zip	3638436	3526370	3519989		

Table 4: Count-flow table separating records, rows, weighted deaths, model-ready cells, and model-comparison cells.

Stage	Years	Unit	Rows/cells	Natality	Linked denom	Infant num	Fetal deaths	Fetal denom
Raw source records	source-specific	records	63884468	33589282	29633197	163804.0	498185	
Singleton valid natality rows	2016-2024	rows	32448118	32448118				
Singleton valid linked denominator rows	2017-2024 period files	rows	28644762		28644762			
Singleton valid linked infant-death rows	2017-2024 period files	rows / weighted deaths	141354			142632.4		
Singleton valid cohort denominator rows	2016-2023 birth cohorts	rows	28872085		28872085			
Singleton valid cohort infant-death rows	2016-2023 birth cohorts	rows / unweighted deaths	143079			143079.0		
Singleton valid fetal-death rows	2014-2024	rows	222075				222075	
Harmonized infant model-ready cells	2016-2023 birth cohorts	aggregate cells	15314		28872085	143077.0		
Harmonized fetal model-ready cells	2016-2024	aggregate cells	19101	32448118			179027	32627145
Infant model-comparison cells	2016-2023 cohorts, 24-42 weeks	aggregate cells	6111		28750979	99253.0		
Fetal model-comparison cells	2016-2024, 24-42 weeks	aggregate cells	7007				96602	32419577

Table 5: Fixed-width fields used for the record-level extraction.

file family	variable	position	role
Natality	sex	475	sex-specific growth modeling
Natality	DPLURAL	454	singleton restriction
Natality	OEGest_Comb	499-500	gestational-age axis
Natality	DBWT	504-507	live-birth birthweight
Linked denominator / linked numerator	sex	475	sex-specific linked-birth modeling
Linked denominator / linked numerator	DPLURAL	454	singleton restriction
Linked denominator / linked numerator	OEGest_Comb	499-500	gestational-age axis

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Table 5: Fixed-width fields used for the record-level extraction.

file family	variable	position	role
Linked denominator / linked numerator	BRTHWGT	512-515	linked-file birthweight
Cohort linked denominator / numerator	DOB_YY	9-12	birth-cohort numerator filtering
Cohort linked denominator / numerator	RESTATUS	104	resident-record filtering for user-created cohort file
Cohort linked denominator / numerator	CO_SEQNUM / CO_YOD	365-371 / 372-375	NCHS cohort linkage identifiers documented for user-created cohort files
Natality/linked birth covariates	MRACEHISP	117	social/demographic stratification
Natality/linked birth covariates	CIG_REC	269	smoking stratification
Natality/linked birth covariates	BMI_R	287	BMI stratification
Linked infant death numerator	AGEDX	1356-1358	age-at-death subgroup
Linked infant death numerator	UCODR130	1373-1375	cause-of-death subgroup
Linked infant death numerator	RECWT	1377-1384	period-linked numerator weighting; creates non-integer death counts
Fetal death	sex	316	sex-specific fetal-death surface
Fetal death	DPLURAL	301	singleton restriction
Fetal death	OEGest_Comb	340-341	gestational-age axis
Fetal death	DBWT	349-352	delivery-weight axis
Fetal death	IICOD	2603-2607	initiating cause of fetal death

Table 6: Linked denominator and natality singleton-count concordance.

Year	Births	Linked births	Difference	Rel diff (%)
2017	3727649	3728351	702	0.019
2018	3669736	3670407	671	0.018
2019	3628705	3629448	743	0.020
2020	3499833	3500729	896	0.026
2021	3547665	3549554	1889	0.053
2022	3553730	3555454	1724	0.049
2023	3487259	3488996	1737	0.050
2024	3519989	3521823	1834	0.052

Table 7: Fitted BCA proxy-equation parameters. N* is birthweight divided by 3500 g; parameters are public-data curve-fit quantities, not direct cellular measurements.

Sex	BCA a	BCA eta	BCA b	BCA c (wk)	BCA t0 (wk)	BCA Wref (g)	MAPE (%)
Female	0.248	1.393	6.052	5.712	40.333	3500.0	0.422
Male	0.334	1.096	6.171	5.818	39.240	3500.0	0.441

Table 8: Temporal out-of-sample benchmark. Cohort infant models validate on 2021-2023 cells and fetal models validate on 2022-2024 cells; validation -2LL is reported instead of AIC.

Outcome	Model	Train years	Validation years	Validation Brier	Validation AUC	Validation -2LL
Cohort infant death	GA only	2016-2020	2021-2023	3.45e-05	0.746	426984.1
Cohort infant death	GA + p10/p90	2016-2020	2021-2023	2.13e-05	0.769	420644.3
Cohort infant death	GA + log residual	2016-2020	2021-2023	2.70e-05	0.777	418700.8
Cohort infant death	GA + BCA residual	2016-2020	2021-2023	2.68e-05	0.777	418569.9
Cohort infant death	GA + BW surface	2016-2020	2021-2023	2.13e-05	0.779	417171.0
Fetal death	GA only	2016-2021	2022-2024	7.27e-05	0.864	311840.8
Fetal death	GA + p10/p90	2016-2021	2022-2024	3.98e-05	0.875	305936.6
Fetal death	GA + log residual	2016-2021	2022-2024	2.43e-05	0.879	302707.0
Fetal death	GA + BCA residual	2016-2021	2022-2024	2.33e-05	0.879	302756.0
Fetal death	GA + BW surface	2016-2021	2022-2024	3.53e-05	0.882	302285.4

Table 9: Temporal validation calibration by denominator-weighted predicted-risk decile. Expected events are predicted risk times denominator within validation cells.

Outcome	Model	Decile	Observed events	Expected events	Denominator	Observed/1000	Predicted/1000	O/E
Cohort infant death	GA + log residual	1.0	1117.0	955.5	1023490	1.091	0.934	1.169
Cohort infant death	GA + log residual	2.0	1275.0	1270.3	1048642	1.216	1.211	1.004
Cohort infant death	GA + log residual	3.0	1268.0	1480.6	1063033	1.193	1.393	0.856
Cohort infant death	GA + log residual	4.0	1487.0	1592.5	1033504	1.439	1.541	0.934
Cohort infant death	GA + log residual	5.0	1602.0	1879.2	1090672	1.469	1.723	0.852
Cohort infant death	GA + log residual	6.0	1881.0	2108.9	1055738	1.782	1.998	0.892
Cohort infant death	GA + log residual	7.0	2156.0	2419.4	1035512	2.082	2.336	0.891
Cohort infant death	GA + log residual	8.0	2840.0	3024.1	1066350	2.663	2.836	0.939
Cohort infant death	GA + log residual	9.0	3840.0	3982.8	1051236	3.653	3.789	0.964
Cohort infant death	GA + log residual	10.0	19441.0	18452.5	1059861	18.343	17.410	1.054
Cohort infant death	GA + BCA residual	1.0	1115.0	928.2	1023528	1.089	0.907	1.201
Cohort infant death	GA + BCA residual	2.0	1317.0	1266.0	1063237	1.239	1.191	1.040
Cohort infant death	GA + BCA residual	3.0	1230.0	1453.2	1054684	1.166	1.378	0.846
Cohort infant death	GA + BCA residual	4.0	1505.0	1583.6	1037142	1.451	1.527	0.950
Cohort infant death	GA + BCA residual	5.0	1579.0	1849.3	1079612	1.463	1.713	0.854
Cohort infant death	GA + BCA residual	6.0	1849.0	2063.4	1036844	1.783	1.990	0.896
Cohort infant death	GA + BCA residual	7.0	2198.0	2477.9	1059297	2.075	2.339	0.887
Cohort infant death	GA + BCA residual	8.0	2811.0	3027.6	1059038	2.654	2.859	0.928
Cohort infant death	GA + BCA residual	9.0	3900.0	4057.7	1051899	3.708	3.858	0.961
Cohort infant death	GA + BCA residual	10.0	19403.0	18494.9	1062757	18.257	17.403	1.049

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Table 9: Temporal validation calibration by denominator-weighted predicted-risk decile. Expected events are predicted risk times denominator within validation cells.

Outcome	Model	Decile	Observed events	Expected events	Denominator	Observed/1000	Predicted/1000	O/E
Cohort infant death	GA + BW surface	1.0	1145.0	800.1	1035725	1.106	0.772	1.431
Cohort infant death	GA + BW surface	2.0	1073.0	1047.8	1032664	1.039	1.015	1.024
Cohort infant death	GA + BW surface	3.0	1304.0	1297.5	1064967	1.224	1.218	1.005
Cohort infant death	GA + BW surface	4.0	1522.0	1464.2	1069221	1.423	1.369	1.039
Cohort infant death	GA + BW surface	5.0	1604.0	1650.8	1043338	1.537	1.582	0.972
Cohort infant death	GA + BW surface	6.0	1922.0	1987.8	1064490	1.806	1.867	0.967
Cohort infant death	GA + BW surface	7.0	2229.0	2335.7	1053019	2.117	2.218	0.954
Cohort infant death	GA + BW surface	8.0	2830.0	2935.6	1058662	2.673	2.773	0.964
Cohort infant death	GA + BW surface	9.0	3841.0	4016.1	1043196	3.682	3.850	0.956
Cohort infant death	GA + BW surface	10.0	19437.0	19277.3	1062756	18.289	18.139	1.008
Fetal death	GA + log residual	1.0	631.0	239.5	1034589	0.610	0.231	2.635
Fetal death	GA + log residual	2.0	598.0	385.4	1053636	0.568	0.366	1.552
Fetal death	GA + log residual	3.0	398.0	576.1	1029691	0.387	0.559	0.691
Fetal death	GA + log residual	4.0	388.0	660.0	1059995	0.366	0.623	0.588
Fetal death	GA + log residual	5.0	468.0	784.0	1093638	0.428	0.717	0.597
Fetal death	GA + log residual	6.0	739.0	1026.0	1050834	0.703	0.976	0.720
Fetal death	GA + log residual	7.0	955.0	1281.0	1042048	0.916	1.229	0.745
Fetal death	GA + log residual	8.0	1451.0	1913.6	1065411	1.362	1.796	0.758
Fetal death	GA + log residual	9.0	2209.0	2951.6	1053267	2.097	2.802	0.748
Fetal death	GA + log residual	10.0	22269.0	23081.1	1067173	20.867	21.628	0.965
Fetal death	GA + BCA residual	1.0	631.0	226.3	1034589	0.610	0.219	2.789
Fetal death	GA + BCA residual	2.0	598.0	366.2	1053636	0.568	0.348	1.633
Fetal death	GA + BCA residual	3.0	398.0	560.7	1029691	0.387	0.544	0.710
Fetal death	GA + BCA residual	4.0	388.0	642.9	1059995	0.366	0.606	0.604
Fetal death	GA + BCA residual	5.0	469.0	764.6	1093702	0.429	0.699	0.613
Fetal death	GA + BCA residual	6.0	742.0	1005.7	1047684	0.708	0.960	0.738
Fetal death	GA + BCA residual	7.0	951.0	1276.3	1045134	0.910	1.221	0.745
Fetal death	GA + BCA residual	8.0	1461.0	1929.9	1066161	1.370	1.810	0.757
Fetal death	GA + BCA residual	9.0	2279.0	3062.0	1064579	2.141	2.876	0.744

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Table 9: Temporal validation calibration by denominator-weighted predicted-risk decile. Expected events are predicted risk times denominator within validation cells.

Outcome	Model	Decile	Observed events	Expected events	Denominator	Observed/1000	Predicted/1000	O/E
Fetal death	GA + BCA residual	10.0	22189.0	23099.2	1055111	21.030	21.893	0.961
Fetal death	GA + BW surface	1.0	582.0	278.3	1054278	0.552	0.264	2.091
Fetal death	GA + BW surface	2.0	575.0	420.0	1034223	0.556	0.406	1.369
Fetal death	GA + BW surface	3.0	340.0	575.5	1037847	0.328	0.554	0.591
Fetal death	GA + BW surface	4.0	390.0	635.0	1047425	0.372	0.606	0.614
Fetal death	GA + BW surface	5.0	516.0	771.5	1085779	0.475	0.711	0.669
Fetal death	GA + BW surface	6.0	691.0	980.0	1048027	0.659	0.935	0.705
Fetal death	GA + BW surface	7.0	980.0	1208.1	1055297	0.929	1.145	0.811
Fetal death	GA + BW surface	8.0	1375.0	1759.8	1074182	1.280	1.638	0.781
Fetal death	GA + BW surface	9.0	2303.0	2624.6	1048051	2.197	2.504	0.877
Fetal death	GA + BW surface	10.0	22354.0	23371.4	1065173	20.986	21.941	0.956

Table 10: Temporal sensitivity refitting expected-size curves on training-period natality data before validating on held-out later years.

Validation design	Outcome	Model	Validation Brier	Validation AUC	Validation -2LL
Cohort infant death train-period expected-size curves	Cohort infant death	GA only	3.45e-05	0.746	426984.1
Cohort infant death train-period expected-size curves	Cohort infant death	GA + p10/p90	2.13e-05	0.769	420644.3
Cohort infant death train-period expected-size curves	Cohort infant death	GA + log residual	2.70e-05	0.777	418695.4
Cohort infant death train-period expected-size curves	Cohort infant death	GA + BCA residual	2.68e-05	0.777	418567.5
Cohort infant death train-period expected-size curves	Cohort infant death	GA + BW surface	2.13e-05	0.779	417171.0
Fetal death train-period expected-size curves	Fetal death	GA only	7.27e-05	0.864	311840.8
Fetal death train-period expected-size curves	Fetal death	GA + p10/p90	3.99e-05	0.875	305943.3

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Table 10: Temporal sensitivity refitting expected-size curves on training-period natality data before validating on held-out later years.

Validation design	Outcome	Model	Validation Brier	Validation AUC	Validation -2LL
Fetal death train-period expected-size curves	Fetal death	GA + log residual	2.43e-05	0.879	302708.6
Fetal death train-period expected-size curves	Fetal death	GA + BCA residual	2.33e-05	0.879	302764.0
Fetal death train-period expected-size curves	Fetal death	GA + BW surface	3.53e-05	0.882	302285.4

Table 11: Leave-one-year-out validation across cohort infant and fetal mortality surfaces.

Outcome	Model	Validation year	Validation Brier	Validation AUC	Validation -2LL
Cohort infant death	GA only	2016	4.03e-05	0.747	152598.9
Cohort infant death	GA + p10/p90	2016	2.79e-05	0.772	150388.4
Cohort infant death	GA + log residual	2016	2.72e-05	0.782	149500.3
Cohort infant death	GA + BCA residual	2016	2.59e-05	0.781	149434.1
Cohort infant death	GA + BW surface	2016	2.66e-05	0.783	148970.9
Cohort infant death	GA only	2017	3.24e-05	0.750	148732.7
Cohort infant death	GA + p10/p90	2017	2.08e-05	0.777	146367.1
Cohort infant death	GA + log residual	2017	2.66e-05	0.784	145669.1
Cohort infant death	GA + BCA residual	2017	2.61e-05	0.784	145614.4
Cohort infant death	GA + BW surface	2017	1.87e-05	0.786	144983.0
Cohort infant death	GA only	2018	3.57e-05	0.749	144412.0
Cohort infant death	GA + p10/p90	2018	2.44e-05	0.774	142142.4
Cohort infant death	GA + log residual	2018	2.60e-05	0.784	141334.5
Cohort infant death	GA + BCA residual	2018	2.56e-05	0.784	141284.6
Cohort infant death	GA + BW surface	2018	2.16e-05	0.784	140743.5
Cohort infant death	GA only	2019	3.81e-05	0.749	140089.3
Cohort infant death	GA + p10/p90	2019	2.44e-05	0.776	137844.6
Cohort infant death	GA + log residual	2019	2.51e-05	0.784	137028.7
Cohort infant death	GA + BCA residual	2019	2.44e-05	0.784	136973.8
Cohort infant death	GA + BW surface	2019	2.47e-05	0.786	136516.1

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Table 11: Leave-one-year-out validation across cohort infant and fetal mortality surfaces.

Outcome	Model	Validation year	Validation Brier	Validation AUC	Validation -2LL
Cohort infant death	GA only	2020	2.88e-05	0.743	134281.0
Cohort infant death	GA + p10/p90	2020	1.98e-05	0.770	132204.3
Cohort infant death	GA + log residual	2020	2.59e-05	0.778	131579.9
Cohort infant death	GA + BCA residual	2020	2.60e-05	0.778	131543.9
Cohort infant death	GA + BW surface	2020	1.69e-05	0.781	130952.5
Cohort infant death	GA only	2021	3.63e-05	0.745	142068.2
Cohort infant death	GA + p10/p90	2021	2.25e-05	0.770	139850.0
Cohort infant death	GA + log residual	2021	2.51e-05	0.777	139149.3
Cohort infant death	GA + BCA residual	2021	2.46e-05	0.777	139093.7
Cohort infant death	GA + BW surface	2021	2.25e-05	0.779	138700.1
Cohort infant death	GA only	2022	3.36e-05	0.742	145498.3
Cohort infant death	GA + p10/p90	2022	1.97e-05	0.766	143424.1
Cohort infant death	GA + log residual	2022	2.45e-05	0.773	142788.3
Cohort infant death	GA + BCA residual	2022	2.41e-05	0.773	142745.5
Cohort infant death	GA + BW surface	2022	2.11e-05	0.774	142300.2
Cohort infant death	GA only	2023	3.36e-05	0.751	139421.3
Cohort infant death	GA + p10/p90	2023	2.18e-05	0.773	137372.2
Cohort infant death	GA + log residual	2023	2.98e-05	0.781	136759.3
Cohort infant death	GA + BCA residual	2023	3.00e-05	0.781	136726.7
Cohort infant death	GA + BW surface	2023	1.99e-05	0.782	136171.2
Fetal death	GA only	2016	1.08e-04	0.866	120991.7
Fetal death	GA + p10/p90	2016	5.04e-05	0.879	117760.0
Fetal death	GA + log residual	2016	1.86e-05	0.883	116076.3
Fetal death	GA + BCA residual	2016	1.77e-05	0.882	116058.9
Fetal death	GA + BW surface	2016	4.55e-05	0.885	116220.2
Fetal death	GA only	2017	1.07e-04	0.869	116083.3
Fetal death	GA + p10/p90	2017	5.06e-05	0.882	112938.9
Fetal death	GA + log residual	2017	2.04e-05	0.887	111322.7
Fetal death	GA + BCA residual	2017	1.93e-05	0.887	111296.0
Fetal death	GA + BW surface	2017	4.79e-05	0.890	111454.1
Fetal death	GA only	2018	1.07e-04	0.870	114361.8
Fetal death	GA + p10/p90	2018	5.03e-05	0.882	111474.7
Fetal death	GA + log residual	2018	1.86e-05	0.887	109860.4
Fetal death	GA + BCA residual	2018	1.74e-05	0.887	109835.9
Fetal death	GA + BW surface	2018	4.93e-05	0.890	110180.3
Fetal death	GA only	2019	1.02e-04	0.869	110173.9

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Table 11: Leave-one-year-out validation across cohort infant and fetal mortality surfaces.

Outcome	Model	Validation year	Validation Brier	Validation AUC	Validation -2LL
Fetal death	GA + p10/p90	2019	4.97e-05	0.882	107476.2
Fetal death	GA + log residual	2019	1.90e-05	0.886	105987.8
Fetal death	GA + BCA residual	2019	1.75e-05	0.886	105948.2
Fetal death	GA + BW surface	2019	4.95e-05	0.889	106244.3
Fetal death	GA only	2020	8.80e-05	0.871	107980.9
Fetal death	GA + p10/p90	2020	4.20e-05	0.884	105443.0
Fetal death	GA + log residual	2020	1.82e-05	0.888	104087.9
Fetal death	GA + BCA residual	2020	1.73e-05	0.888	104075.4
Fetal death	GA + BW surface	2020	3.74e-05	0.891	104038.0
Fetal death	GA only	2021	8.10e-05	0.872	109707.5
Fetal death	GA + p10/p90	2021	4.43e-05	0.883	107520.0
Fetal death	GA + log residual	2021	2.30e-05	0.888	106247.6
Fetal death	GA + BCA residual	2021	2.18e-05	0.887	106245.2
Fetal death	GA + BW surface	2021	3.73e-05	0.890	106129.8
Fetal death	GA only	2022	7.60e-05	0.860	107789.4
Fetal death	GA + p10/p90	2022	4.07e-05	0.873	105564.6
Fetal death	GA + log residual	2022	2.29e-05	0.877	104365.8
Fetal death	GA + BCA residual	2022	2.20e-05	0.877	104377.6
Fetal death	GA + BW surface	2022	3.52e-05	0.881	104137.2
Fetal death	GA only	2023	8.14e-05	0.867	102266.0
Fetal death	GA + p10/p90	2023	4.02e-05	0.877	100194.2
Fetal death	GA + log residual	2023	2.16e-05	0.881	99110.2
Fetal death	GA + BCA residual	2023	2.06e-05	0.881	99124.4
Fetal death	GA + BW surface	2023	3.91e-05	0.884	99128.6
Fetal death	GA only	2024	5.99e-05	0.865	101640.0
Fetal death	GA + p10/p90	2024	3.66e-05	0.876	100007.3
Fetal death	GA + log residual	2024	2.35e-05	0.879	99058.3
Fetal death	GA + BCA residual	2024	2.25e-05	0.879	99073.8
Fetal death	GA + BW surface	2024	2.98e-05	0.882	98900.4

Table 12: Temporal validation decision-curve style net-benefit summary at selected risk thresholds.

Outcome	Model	Threshold/1000	Flagged denominator	Flagged events	Net benefit	NB - treat all
Cohort infant death	GA + p10/p90	1.000	9953503	36147.0	0.002	-0.000
Cohort infant death	GA + p10/p90	2.500	2711700	24755.0	0.002	0.001
Cohort infant death	GA + p10/p90	5.000	1126927	19191.0	0.001	0.003
Cohort infant death	GA + p10/p90	10.000	436177	14713.0	0.001	0.008
Cohort infant death	GA + p10/p90	20.000	156268	10546.0	0.001	0.018
Cohort infant death	GA + log residual	1.000	9807047	36100.0	0.002	-0.000
Cohort infant death	GA + log residual	2.500	3301446	26393.0	0.002	0.001
Cohort infant death	GA + log residual	5.000	968122	18969.0	0.001	0.003

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Table 12: Temporal validation decision-curve style net-benefit summary at selected risk thresholds.

Outcome	Model	Threshold/1000	Flagged denominator	Flagged events	Net benefit	NB - treat all
Cohort infant death	GA + log residual	10.000	332634	14075.0	0.001	0.008
Cohort infant death	GA + log residual	20.000	147095	10669.0	0.001	0.018
Cohort infant death	GA + BCA residual	1.000	9754245	36065.0	0.003	-0.000
Cohort infant death	GA + BCA residual	2.500	3303406	26397.0	0.002	0.001
Cohort infant death	GA + BCA residual	5.000	976552	19031.0	0.001	0.003
Cohort infant death	GA + BCA residual	10.000	335372	14115.0	0.001	0.008
Cohort infant death	GA + BCA residual	20.000	145809	10658.0	0.001	0.018
Cohort infant death	GA + BW surface	1.000	8910891	35184.0	0.002	-0.000
Cohort infant death	GA + BW surface	2.500	3136015	26058.0	0.002	0.001
Cohort infant death	GA + BW surface	5.000	1015676	19223.0	0.001	0.003
Cohort infant death	GA + BW surface	10.000	379448	14677.0	0.001	0.008
Cohort infant death	GA + BW surface	20.000	175792	11547.0	0.001	0.018
Fetal death	GA + p10/p90	1.000	4461401	27179.0	0.002	0.000
Fetal death	GA + p10/p90	2.500	1937246	23970.0	0.002	0.001
Fetal death	GA + p10/p90	5.000	783902	20863.0	0.002	0.004
Fetal death	GA + p10/p90	10.000	398950	18420.0	0.001	0.009
Fetal death	GA + p10/p90	20.000	238618	16222.0	0.001	0.019
Fetal death	GA + log residual	1.000	4815734	27323.0	0.002	0.000
Fetal death	GA + log residual	2.500	1707116	23855.0	0.002	0.002
Fetal death	GA + log residual	5.000	782841	21261.0	0.002	0.004
Fetal death	GA + log residual	10.000	397916	18573.0	0.001	0.009
Fetal death	GA + log residual	20.000	218925	15888.0	0.001	0.019
Fetal death	GA + BCA residual	1.000	4814090	27318.0	0.002	0.000
Fetal death	GA + BCA residual	2.500	1748999	23920.0	0.002	0.002
Fetal death	GA + BCA residual	5.000	835425	21445.0	0.002	0.004
Fetal death	GA + BCA residual	10.000	416636	18762.0	0.001	0.009
Fetal death	GA + BCA residual	20.000	225785	16082.0	0.001	0.019
Fetal death	GA + BW surface	1.000	4554985	27243.0	0.002	0.000
Fetal death	GA + BW surface	2.500	1525219	23714.0	0.002	0.002
Fetal death	GA + BW surface	5.000	727641	20983.0	0.002	0.004
Fetal death	GA + BW surface	10.000	385747	18445.0	0.001	0.009
Fetal death	GA + BW surface	20.000	221504	15935.0	0.001	0.019

Table 13: In-sample sensitivity adding calendar-year fixed effects. This checks whether benchmark patterns persist after absorbing year-to-year shifts.

Outcome	Model	AIC	Brier	Weighted AUC	Pseudo R2
Cohort infant death	GA only + year FE	959600.4	3.47e-05	0.748	0.817
Cohort infant death	GA + p10/p90 + year FE	942100.7	2.25e-05	0.773	0.898
Cohort infant death	GA + log residual + year FE	936323.6	2.61e-05	0.780	0.924
Cohort infant death	GA + BCA residual + year FE	935928.8	2.56e-05	0.780	0.926
Cohort infant death	GA + BW surface + year FE	931864.8	2.14e-05	0.782	0.945
Fetal death	GA only + year FE	815316.1	9.07e-05	0.869	0.855
Fetal death	GA + p10/p90 + year FE	792690.5	4.48e-05	0.880	0.914
Fetal death	GA + log residual + year FE	780436.4	2.03e-05	0.884	0.946
Fetal death	GA + BCA residual + year FE	780345.5	1.92e-05	0.884	0.947
Fetal death	GA + BW surface + year FE	780842.9	4.12e-05	0.887	0.945

Table 14: RECORD/STROBE-style reporting checklist for public routinely collected data.

Reporting domain	Location	Audit response
Data source	Data; Supplementary Tables 1-4	Public NCHS natality, period/cohort linked birth-infant death, fetal-death files, CDC WONDER exports, and public fetal-growth standards are listed with local paths and roles.
Linkage and aggregation	Methods; Count-flow table	Records are not treated as private individual data; they are collapsed into year/sex/gestational-age/100-g aggregate cells before outcome modeling.
Eligibility	Methods; Supplementary field map	Primary analyses restrict to singleton records with usable sex, obstetric-estimate gestational age, and birthweight or delivery weight.
Cohort-linked design	Methods; Limitations	Primary infant mortality uses user-created birth-cohort linked cells. Period-linked weighted RECWT analyses are retained only as sensitivity checks.
Fetal-death denominator	Methods; Limitations	Fetal surfaces use fetal deaths divided by fetal deaths plus matched live births and are described as delivery-based mortality surfaces, not prospective hazards.
Sparse cells and privacy	Methods; Limitations	Model comparisons restrict to cells with denominator >=50 and exclude internally inconsistent weighted cells; CDC WONDER suppression/privacy limits are not treated as individual evidence.

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Table 14: RECORD/STROBE-style reporting checklist for public routinely collected data.

Reporting domain	Location	Audit response
Code and provenance	Ethics/Data Availability; Supplement; Audit trail	Scripts, processed CSVs, source inventory, SHA-256 hashes, ZIP integrity checks, and regeneration commands are packaged with the deliverable.

Table 15: Birth-cohort linked primary infant endpoint compared with period-linked weighted sensitivity endpoint.

Endpoint	Years	Growth zone	Cells	Deaths	Denominator	Rate/1000	RR vs p10-p90
Birth-cohort linked primary	2016-2023 cohorts	Above empirical p90	6615	7350.0	2347654	3.131	1.025
Birth-cohort linked primary	2016-2023 cohorts	Below empirical p10	3306	23821.0	2263899	10.522	3.445
Birth-cohort linked primary	2016-2023 cohorts	Empirical p10-p90	2928	73931.0	24204382	3.054	1.000
Period-linked weighted sensitivity	2017-2024 periods	Above empirical p90	6599	7308.6	2319335	3.151	1.026
Period-linked weighted sensitivity	2017-2024 periods	Below empirical p10	3294	23786.8	2231235	10.661	3.470
Period-linked weighted sensitivity	2017-2024 periods	Empirical p10-p90	2928	73849.4	24038854	3.072	1.000

Table 16: Infant-death growth-zone sensitivity to excluding period-linked cells where weighted infant deaths exceed same-cell linked denominator births.

Scenario	Growth zone	Cells	Deaths	Denominator	Rate/1000	RR vs p10-p90
Period-linked weighted sensitivity table	Above empirical p90	6599	7308.6	2319335	3.151	1.026
Period-linked weighted sensitivity table	Below empirical p10	3294	23786.8	2231235	10.661	3.470
Period-linked weighted sensitivity table	Empirical p10-p90	2928	73849.4	24038854	3.072	1.000
Exclude weighted deaths > linked denominator births	Above empirical p90	6540	7235.9	2319267	3.120	1.016
Exclude weighted deaths > linked denominator births	Below empirical p10	3211	23617.8	2231076	10.586	3.446

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Table 16: Infant-death growth-zone sensitivity to excluding period-linked cells where weighted infant deaths exceed same-cell linked denominator births.

Scenario	Growth zone	Cells	Deaths	Denominator	Rate/1000	RR vs p10-p90
Exclude weighted deaths > linked denominator births	Empirical p10-p90	2928	73849.4	24038854	3.072	1.000

Table 17: Mortality rates by compact residual bin.

Outcome	Residual bin	Deaths	Denominator	Rate/1000
Cohort infant death	< -50%	2729.0	13910	196.190
Cohort infant death	-50 to -30%	11830.0	254284	46.523
Cohort infant death	-30 to -20%	11783.0	1316179	8.952
Cohort infant death	-20 to -10%	19459.0	4620109	4.212
Cohort infant death	-10 to +10%	41274.0	16850890	2.449
Cohort infant death	+10 to +20%	9054.0	3983064	2.273
Cohort infant death	+20 to +30%	4303.0	1230386	3.497
Cohort infant death	+30 to +50%	2937.0	480423	6.113
Cohort infant death	> +50%	1733.0	66690	25.986
Fetal death	< -50%	10644.0	26088	408.004
Fetal death	-50 to -30%	17088.0	301461	56.684
Fetal death	-30 to -20%	13668.0	1490054	9.173
Fetal death	-20 to -10%	18842.0	5218025	3.611
Fetal death	-10 to +10%	31346.0	18972911	1.652
Fetal death	+10 to +20%	7129.0	4483355	1.590
Fetal death	+20 to +30%	3911.0	1386920	2.820
Fetal death	+30 to +50%	3706.0	543183	6.823
Fetal death	> +50%	2802.0	77810	36.011

Table 18: Descriptive BMI-group mortality rates and rate ratios.

Outcome	Covariate	Level	Deaths	Denominator	Rate/1000	RR vs reference
Cohort infant death	bmi_group	Unknown	8823.0	642689	13.728	3.470
Cohort infant death	bmi_group	Obese III	11883.0	1583859	7.503	1.897
Cohort infant death	bmi_group	Obese II	13914.0	2204126	6.313	1.596
Cohort infant death	bmi_group	Obese I	24053.0	4430363	5.429	1.372
Cohort infant death	bmi_group	Underweight	4598.0	855744	5.373	1.358
Cohort infant death	bmi_group	Overweight	34138.0	7610042	4.486	1.134
Cohort infant death	bmi_group	Normal	45670.0	11545262	3.956	1.000
Fetal death	bmi_group	Unknown	22173.0	748231	29.634	7.619
Fetal death	bmi_group	Obese III	14536.0	1811467	8.024	2.063
Fetal death	bmi_group	Obese II	16989.0	2517005	6.750	1.735
Fetal death	bmi_group	Obese I	29518.0	5056044	5.838	1.501
Fetal death	bmi_group	Overweight	41302.0	8632278	4.785	1.230
Fetal death	bmi_group	Underweight	4301.0	952719	4.514	1.161
Fetal death	bmi_group	Normal	50208.0	12909401	3.889	1.000

Table 19: Descriptive diabetes, hypertension, and smoking mortality rates and rate ratios.

Outcome	Covariate	Level	Deaths	Denominator	Rate/1000	RR vs reference
Cohort infant death	diabetes	Pregestational diabetes	3240.0	295065	10.981	2.202
Cohort infant death	diabetes	No diabetes	132031.0	26471750	4.988	1.000
Cohort infant death	diabetes	Gestational diabetes	7248.0	2076264	3.491	0.700
Cohort infant death	hypertension	Prep pregnancy hypertension	6378.0	677597	9.413	1.969
Cohort infant death	hypertension	Gestational hypertension	11987.0	2252810	5.321	1.113
Cohort infant death	hypertension	No hypertension	123575.0	25855804	4.779	1.000
Cohort infant death	smoking	Yes	15953.0	1568205	10.173	2.212
Cohort infant death	smoking	No	124989.0	27178735	4.599	1.000
Fetal death	diabetes	Pregestational diabetes	6932.0	347945	19.923	3.852
Fetal death	diabetes	No diabetes	154397.0	29853256	5.172	1.000
Fetal death	diabetes	Gestational diabetes	7852.0	2382175	3.296	0.637
Fetal death	hypertension	Prep pregnancy hypertension	10737.0	804382	13.348	2.662
Fetal death	hypertension	No hypertension	145842.0	29090102	5.013	1.000
Fetal death	hypertension	Gestational hypertension	11823.0	2625490	4.503	0.898
Fetal death	smoking	Yes	14595.0	1665835	8.761	1.793
Fetal death	smoking	No	150498.0	30806727	4.885	1.000

Table 20: Descriptive race/ethnicity-stratified mortality rates and rate ratios. Race/ethnicity is treated as social and clinical stratification, not genetic explanation.

Outcome	Covariate	Level	Deaths	Denominator	Rate/1000	RR vs reference
Cohort infant death	race_hisp	Other	2642.0	258817	10.208	2.544
Cohort infant death	race_hisp	NH Black	38783.0	4087182	9.489	2.365
Cohort infant death	race_hisp	NH AIAN	1674.0	215817	7.757	1.933
Cohort infant death	race_hisp	NH multirace	3813.0	656207	5.811	1.448
Cohort infant death	race_hisp	Hispanic	30885.0	7040008	4.387	1.093
Cohort infant death	race_hisp	NH White	59142.0	14738718	4.013	1.000
Cohort infant death	race_hisp	NH Asian	5574.0	1800052	3.097	0.772
Fetal death	race_hisp	Other	6050.0	299260	20.217	4.467
Fetal death	race_hisp	NH Black	46857.0	4586970	10.215	2.257
Fetal death	race_hisp	NH AIAN	1659.0	240832	6.889	1.522
Fetal death	race_hisp	Hispanic	37168.0	8093977	4.592	1.015
Fetal death	race_hisp	NH White	74850.0	16540118	4.525	1.000
Fetal death	race_hisp	NH multirace	2848.0	747597	3.810	0.842
Fetal death	race_hisp	NH Asian	7579.0	2030956	3.732	0.825

Table 21: WHO and INTERGROWTH EFW medians compared with U.S. birthweight medians.

GA week	US BW p10	US BW p50	US BW p90	WHO EFW p10	WHO EFW p50	WHO EFW p90	IG EFW p10	IG EFW p50	IG EFW p90	WHO-US p50	IG-US p50
24	500.0	650.0	819.1	576.0	665.0	765.0	559.3	629.9	713.8	15.0	-20.1
28	800.0	1122.8	1450.0	1026.0	1189.0	1368.0	982.4	1119.3	1279.5	66.2	-3.6
32	1400.0	1800.0	2300.0	1635.0	1901.0	2187.0	1555.5	1794.8	2071.7	101.0	-5.2
36	2300.0	2800.0	3400.0	2352.0	2745.0	3153.0	2228.3	2602.2	3030.5	-55.0	-197.8
40	3000.0	3500.0	4050.0	3084.0	3617.0	4135.0	2888.4	3408.6	3999.3	117.0	-91.4

Table 22: WHO, INTERGROWTH, and NICHD biometry standards at selected gestational ages.

Standard	Group	Measure	GA week	Value
INTERGROWTH-21st	Global	AC	20	147.7
NICHD	Non-Hispanic White	AC	20	151.7
WHO	Global	AC	20	150.0
INTERGROWTH-21st	Global	AC	24	191.2
NICHD	Non-Hispanic White	AC	24	198.4
WHO	Global	AC	24	197.0
INTERGROWTH-21st	Global	AC	28	233.3
NICHD	Non-Hispanic White	AC	28	242.3
WHO	Global	AC	28	240.0
INTERGROWTH-21st	Global	AC	32	273.9
NICHD	Non-Hispanic White	AC	32	286.5
WHO	Global	AC	32	279.0
INTERGROWTH-21st	Global	AC	36	312.8
NICHD	Non-Hispanic White	AC	36	328.8
WHO	Global	AC	36	317.0
INTERGROWTH-21st	Global	AC	40	349.8
NICHD	Non-Hispanic White	AC	40	361.4
WHO	Global	AC	40	363.0
INTERGROWTH-21st	Global	BPD	20	48.4
NICHD	Non-Hispanic White	BPD	20	46.3
WHO	Global	BPD	20	47.0
INTERGROWTH-21st	Global	BPD	24	61.4
NICHD	Non-Hispanic White	BPD	24	58.9
WHO	Global	BPD	24	60.0
INTERGROWTH-21st	Global	BPD	28	73.5
NICHD	Non-Hispanic White	BPD	28	70.8
WHO	Global	BPD	28	72.0
INTERGROWTH-21st	Global	BPD	32	83.8
NICHD	Non-Hispanic White	BPD	32	81.3
WHO	Global	BPD	32	81.0
INTERGROWTH-21st	Global	BPD	36	91.2
NICHD	Non-Hispanic White	BPD	36	89.0
WHO	Global	BPD	36	89.0

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Table 22: WHO, INTERGROWTH, and NICHD biometry standards at selected gestational ages.

Standard	Group	Measure	GA week	Value
INTERGROWTH-21st	Global	BPD	40	94.9
NICHD	Non-Hispanic White	BPD	40	93.3
WHO	Global	BPD	40	96.0
INTERGROWTH-21st	Global	FL	20	31.3
NICHD	Non-Hispanic White	FL	20	31.7
WHO	Global	FL	20	32.0
INTERGROWTH-21st	Global	FL	24	41.9
NICHD	Non-Hispanic White	FL	24	42.7
WHO	Global	FL	24	43.0
INTERGROWTH-21st	Global	FL	28	51.3
NICHD	Non-Hispanic White	FL	28	52.0
WHO	Global	FL	28	52.0
INTERGROWTH-21st	Global	FL	32	59.4
NICHD	Non-Hispanic White	FL	32	60.4
WHO	Global	FL	32	61.0
INTERGROWTH-21st	Global	FL	36	66.4
NICHD	Non-Hispanic White	FL	36	67.7
WHO	Global	FL	36	69.0
INTERGROWTH-21st	Global	FL	40	72.1
NICHD	Non-Hispanic White	FL	40	72.8
WHO	Global	FL	40	73.0
INTERGROWTH-21st	Global	HC	20	172.5
NICHD	Non-Hispanic White	HC	20	172.2
WHO	Global	HC	20	173.0
INTERGROWTH-21st	Global	HC	24	219.1
NICHD	Non-Hispanic White	HC	24	219.4
WHO	Global	HC	24	222.0
INTERGROWTH-21st	Global	HC	28	260.4
NICHD	Non-Hispanic White	HC	28	262.7
WHO	Global	HC	28	264.0
INTERGROWTH-21st	Global	HC	32	294.4
NICHD	Non-Hispanic White	HC	32	298.5
WHO	Global	HC	32	296.0
INTERGROWTH-21st	Global	HC	36	319.4
NICHD	Non-Hispanic White	HC	36	323.5
WHO	Global	HC	36	321.0
INTERGROWTH-21st	Global	HC	40	333.9
NICHD	Non-Hispanic White	HC	40	338.0
WHO	Global	HC	40	342.0

Table 23: Aggregate-bin correlations between residual metrics and mortality rates.

Outcome	Metric	Cells	Pearson	Spearman	Weighted Pearson
Cohort infant death	relative_residual	12849	-0.136	-0.438	-0.121
Cohort infant death	abs_relative_residual	12849	0.087	-0.317	0.206
Cohort infant death	bca_relative_residual	12849	-0.142	-0.435	-0.123
Cohort infant death	bca_abs_relative_residual	12849	0.082	-0.315	0.208
Cohort infant death	weight_mid_g	12849	-0.349	-0.587	-0.413
Fetal death	relative_residual	15292	-0.323	-0.492	-0.150
Fetal death	abs_relative_residual	15292	0.163	-0.144	0.257
Fetal death	bca_relative_residual	15292	-0.330	-0.492	-0.148
Fetal death	bca_abs_relative_residual	15292	0.156	-0.149	0.257
Fetal death	weight_mid_g	15292	-0.489	-0.667	-0.403

Table 24: Cohort linked infant deaths by age-at-death group.

Age at death	Cohort infant deaths
Early neonatal (0-6 d)	71111.0
Late neonatal (7-27 d)	19280.0
Postneonatal (28-364 d)	52688.0

Table 25: Top cohort linked infant-death 130-cause recodes.

Infant cause recode	Cohort infant deaths
089	15526.0
135	10422.0
136	8952.0
117	8423.0
146	7013.0
123	6881.0
090	4577.0
076	3988.0
106	3794.0
128	3669.0
130	3411.0
132	3283.0
127	3002.0
080	2665.0
096	2558.0

Table 26: Top initiating fetal-death ICD-10 codes.

Fetal cause code	Fetal deaths
P95	77643.0
P021	16849.0
P011	15856.0
Unknown	13120.0
P025	11321.0

Continued on next page

Table 26: Top initiating fetal-death ICD-10 codes.

Fetal cause code	Fetal deaths
P000	9876.0
P022	9359.0
P027	5422.0
P008	4777.0
P026	4679.0
P010	3812.0
P701	3428.0
Q913	2833.0
P072	2630.0
P832	2263.0

Table 27: Cause-group-specific perinatal mortality rates (per 1000) by empirical birthweight-for-gestational-age growth zone, gestational weeks 24-42. Death counts are deterministic tallies of NCHS linked-infant-death and fetal-death records mapped to ICD-10-based cause groups.

endpoint	growth zone	cause group	deaths	zone denominator	rate per 1000
Cohort infant death	Above empirical p90	Asphyxia/hypoxia	180	2347654.0000	0.0767
Cohort infant death	Above empirical p90	Congenital anomalies	2061	2347654.0000	0.8779
Cohort infant death	Above empirical p90	Haematologic/hydrops/IDM	982	2347654.0000	0.4183
Cohort infant death	Above empirical p90	Infection	408	2347654.0000	0.1738
Cohort infant death	Above empirical p90	Maternal complications	198	2347654.0000	0.0843
Cohort infant death	Above empirical p90	Other perinatal	533	2347654.0000	0.2270
Cohort infant death	Above empirical p90	Other/residual	1443	2347654.0000	0.6147
Cohort infant death	Above empirical p90	Perinatal respiratory	286	2347654.0000	0.1218
Cohort infant death	Above empirical p90	Placental/cord/membrane	143	2347654.0000	0.0609
Cohort infant death	Above empirical p90	Prematurity/growth-related	126	2347654.0000	0.0537
Cohort infant death	Above empirical p90	SIDS/SUID/ill-defined	990	2347654.0000	0.4217
Cohort infant death	Below empirical p10	Asphyxia/hypoxia	266	2263899.0000	0.1175
Cohort infant death	Below empirical p10	Congenital anomalies	11168	2263899.0000	4.9331
Cohort infant death	Below empirical p10	Haematologic/hydrops/IDM	617	2263899.0000	0.2725
Cohort infant death	Below empirical p10	Infection	1252	2263899.0000	0.5530
Cohort infant death	Below empirical p10	Maternal complications	373	2263899.0000	0.1648
Cohort infant death	Below empirical p10	Other perinatal	928	2263899.0000	0.4099
Cohort infant death	Below empirical p10	Other/residual	3819	2263899.0000	1.6869
Cohort infant death	Below empirical p10	Perinatal respiratory	1087	2263899.0000	0.4801
Cohort infant death	Below empirical p10	Placental/cord/membrane	217	2263899.0000	0.0959
Cohort infant death	Below empirical p10	Prematurity/growth-related	1144	2263899.0000	0.5053

endpoint	growth zone	cause group	deaths	zone denominator	rate per 1000
Cohort infant death	Below empirical p10	SIDS/SUID/ill-defined	2950	2263899.0000	1.3031
Cohort infant death	Empirical p10-p90	Asphyxia/hypoxia	1341	24204382.0000	0.0554
Cohort infant death	Empirical p10-p90	Congenital anomalies	16725	24204382.0000	0.6910
Cohort infant death	Empirical p10-p90	Haematologic/hydrops/IDM	3701	24204382.0000	0.1530
Cohort infant death	Empirical p10-p90	Infection	6266	24204382.0000	0.2589
Cohort infant death	Empirical p10-p90	Maternal complications	1864	24204382.0000	0.0770
Cohort infant death	Empirical p10-p90	Other perinatal	4155	24204382.0000	0.1717
Cohort infant death	Empirical p10-p90	Other/residual	17147	24204382.0000	0.7084
Cohort infant death	Empirical p10-p90	Perinatal respiratory	3919	24204382.0000	0.1619
Cohort infant death	Empirical p10-p90	Placental/cord/membrane	1509	24204382.0000	0.0623
Cohort infant death	Empirical p10-p90	Prematurity/growth-related	2030	24204382.0000	0.0839
Cohort infant death	Empirical p10-p90	SIDS/SUID/ill-defined	15271	24204382.0000	0.6309
Fetal death	Above empirical p90	Asphyxia/hypoxia	12	2641083.0000	0.0045
Fetal death	Above empirical p90	Congenital anomalies	692	2641083.0000	0.2620
Fetal death	Above empirical p90	Haematologic/hydrops/IDM	1754	2641083.0000	0.6641
Fetal death	Above empirical p90	Infection	11	2641083.0000	0.0042
Fetal death	Above empirical p90	Maternal complications	1334	2641083.0000	0.5051
Fetal death	Above empirical p90	Other perinatal	38	2641083.0000	0.0144
Fetal death	Above empirical p90	Other/residual	112	2641083.0000	0.0424
Fetal death	Above empirical p90	Perinatal respiratory	10	2641083.0000	0.0038
Fetal death	Above empirical p90	Placental/cord/membrane	1499	2641083.0000	0.5676
Fetal death	Above empirical p90	Prematurity/growth-related	78	2641083.0000	0.0295
Fetal death	Above empirical p90	Unspecified/ill-defined	3041	2641083.0000	1.1514
Fetal death	Below empirical p10	Asphyxia/hypoxia	30	2569428.0000	0.0117
Fetal death	Below empirical p10	Congenital anomalies	4774	2569428.0000	1.8580
Fetal death	Below empirical p10	Haematologic/hydrops/IDM	767	2569428.0000	0.2985
Fetal death	Below empirical p10	Infection	29	2569428.0000	0.0113
Fetal death	Below empirical p10	Maternal complications	5680	2569428.0000	2.2106
Fetal death	Below empirical p10	Other perinatal	102	2569428.0000	0.0397
Fetal death	Below empirical p10	Other/residual	135	2569428.0000	0.0525
Fetal death	Below empirical p10	Perinatal respiratory	19	2569428.0000	0.0074
Fetal death	Below empirical p10	Placental/cord/membrane	7434	2569428.0000	2.8933
Fetal death	Below empirical p10	Prematurity/growth-related	1191	2569428.0000	0.4635
Fetal death	Below empirical p10	Unspecified/ill-defined	12945	2569428.0000	5.0381
Fetal death	Empirical p10-p90	Asphyxia/hypoxia	86	27289296.0000	0.0032

endpoint	growth zone	cause group	deaths	zone denominator	rate per 1000
Fetal death	Empirical p10-p90	Congenital anomalies	4348	27289296.0000	0.1593
Fetal death	Empirical p10-p90	Haematologic/hydrops/SUDM	2528	27289296.0000	0.0925
Fetal death	Empirical p10-p90	Infection	61	27289296.0000	0.0022
Fetal death	Empirical p10-p90	Maternal complications	11004	27289296.0000	0.4032
Fetal death	Empirical p10-p90	Other perinatal	246	27289296.0000	0.0090
Fetal death	Empirical p10-p90	Other/residual	312	27289296.0000	0.0114
Fetal death	Empirical p10-p90	Perinatal respiratory	33	27289296.0000	0.0012
Fetal death	Empirical p10-p90	Placental/cord/membrane	20513	27289296.0000	0.7517
Fetal death	Empirical p10-p90	Prematurity/growth-related	813	27289296.0000	0.0298
Fetal death	Empirical p10-p90	Unspecified/ill-defined	27510	27289296.0000	1.0081

Table 28: Coverage check: total cause-attributed deaths (weeks 24-42) versus the growth-mortality surface death totals, confirming the cause re-parse reconciles with the primary surfaces.

endpoint	cause parsed deaths 24 42wk	surface deaths 24 42wk	coverage fraction	n cause cells
Cohort infant death	105102.0000	105102.0000	1.0000	69878.0000
Fetal death	109136.0000	109136.0000	1.0000	86789.0000

Table 29: Per-growth-zone contributions to the Black-White perinatal mortality gap, with race-specific zone shares and zone-specific mortality rates.

endpoint	growth zone	white share	black share	white rate per 1000	black rate per 1000	distribution contribution	rate contribution
Cohort infant death	Below empirical p10	0.0619	0.1295	11.1649	12.0412	0.7842	0.0839
Cohort infant death	Empirical p10-p90	0.8415	0.8205	2.6845	5.5164	-0.0862	2.3534
Cohort infant death	Above empirical p90	0.0965	0.0500	2.6428	5.9702	-0.2005	0.2438
Fetal death	Below empirical p10	0.0622	0.1308	16.3224	18.6378	1.1987	0.2235
Fetal death	Empirical p10-p90	0.8413	0.8190	2.6124	5.3369	-0.0886	2.2619
Fetal death	Above empirical p90	0.0964	0.0502	2.8144	9.7320	-0.2903	0.5070

Table 30: Period-specific growth-zone perinatal mortality rate ratios (versus the empirical p10-p90 zone) for pre-pandemic, pandemic, and post-pandemic periods, with delta-method 95% confidence intervals.

outcome	period	growth zone	deaths	denominator	rate per 1000	rate ratio vs reference	rr 95ci lower	rr 95ci upper	log rr	log rr se
Cohort infant death	Pre-pandemic (2016-2019)	Below empirical p10	12237.0000	1175196.0000	10.4127	3.4052	3.3368	3.4750	1.2253	0.0104

outcome	period	growth zone	deaths	denominator	rate per 1000	rate ratio vs reference	rr 95ci lower	rr 95ci upper	log rr	log rr se
Cohort infant death	Pre-pandemic (2016-2019)	Above empirical p90	3783.0000	1210130.0000	3.1261	1.0223	0.9887	1.0570	0.0221	0.0170
Cohort infant death	Pandemic (2020-2021)	Below empirical p10	5687.0000	542383.0000	10.4852	3.4983	3.3957	3.6039	1.2523	0.0152
Cohort infant death	Pandemic (2020-2021)	Above empirical p90	1726.0000	578968.0000	2.9812	0.9946	0.9467	1.0449	-0.0054	0.0252
Cohort infant death	Post-pandemic (2022-2023)	Below empirical p10	5897.0000	546320.0000	10.7940	3.4772	3.3771	3.5802	1.2462	0.0149
Cohort infant death	Post-pandemic (2022-2023)	Above empirical p90	1841.0000	558556.0000	3.2960	1.0618	1.0122	1.1138	0.0599	0.0244
Fetal death	Pre-pandemic (2016-2019)	Below empirical p10	15995.0000	1194389.0000	13.3918	5.3919	5.2904	5.4953	1.6849	0.0097
Fetal death	Pre-pandemic (2016-2019)	Above empirical p90	3939.0000	1216248.0000	3.2386	1.3040	1.2615	1.3479	0.2654	0.0169
Fetal death	Pandemic (2020-2021)	Below empirical p10	7128.0000	550301.0000	12.9529	5.0896	4.9489	5.2343	1.6272	0.0143
Fetal death	Pandemic (2020-2021)	Above empirical p90	1860.0000	581315.0000	3.1996	1.2572	1.1982	1.3192	0.2289	0.0245
Fetal death	Post-pandemic (2022-2024)	Below empirical p10	9983.0000	824738.0000	12.1045	5.0311	4.9135	5.1514	1.6156	0.0121
Fetal death	Post-pandemic (2022-2024)	Above empirical p90	2782.0000	843520.0000	3.2981	1.3708	1.3178	1.4260	0.3154	0.0201

Table 31: Ratio-of-rate-ratios shift tests comparing pandemic and post-pandemic growth-zone rate ratios to the pre-pandemic baseline. A ratio near 1 with a confidence interval spanning 1 indicates the growth-mortality relationship was stable through the pandemic shock.

outcome	growth zone	comparison	ror	ror 95ci lower	ror 95ci upper	z statistic	p value
Cohort infant death	Below empirical p10	Pandemic (2020-2021) vs Pre-pandemic (2016-2019)	1.0273	0.9910	1.0650	1.4681	0.1421
Cohort infant death	Below empirical p10	Post-pandemic (2022-2023) vs Pre-pandemic (2016-2019)	1.0211	0.9855	1.0581	1.1530	0.2489
Cohort infant death	Above empirical p90	Pandemic (2020-2021) vs Pre-pandemic (2016-2019)	0.9729	0.9167	1.0327	-0.9028	0.3666

outcome	growth zone	comparison	ror	ror 95ci lower	ror 95ci upper	z statistic	p value
Cohort infant death	Above empirical p90	Post-pandemic (2022-2023) vs Pre-pandemic (2016-2019)	1.0386	0.9798	1.1010	1.2728	0.2031
Fetal death	Below empirical p10	Pandemic (2020-2021) vs Pre-pandemic (2016-2019)	0.9439	0.9125	0.9765	-3.3392	0.0008
Fetal death	Below empirical p10	Post-pandemic (2022-2024) vs Pre-pandemic (2016-2019)	0.9331	0.9052	0.9618	-4.4757	0.0000
Fetal death	Above empirical p90	Pandemic (2020-2021) vs Pre-pandemic (2016-2019)	0.9642	0.9095	1.0221	-1.2250	0.2206
Fetal death	Above empirical p90	Post-pandemic (2022-2024) vs Pre-pandemic (2016-2019)	1.0513	0.9985	1.1068	1.9032	0.0570

Table 32: Weighted AUC of a GA + log-residual grouped-binomial model trained on the pre-pandemic period and evaluated within each period, testing whether the pre-pandemic growth-mortality model continued to discriminate risk.

outcome	period	model	weighted auc	n cells	total deaths
Cohort infant death	Pre-pandemic (2016-2019)	GA + log residual (trained pre-pandemic)	0.7833	3078	50890.0000
Cohort infant death	Pandemic (2020-2021)	GA + log residual (trained pre-pandemic)	0.7775	1518	23709.0000
Cohort infant death	Post-pandemic (2022-2023)	GA + log residual (trained pre-pandemic)	0.7768	1515	24654.0000
Fetal death	Pre-pandemic (2016-2019)	GA + log residual (trained pre-pandemic)	0.8860	3145	45136.0000
Fetal death	Pandemic (2020-2021)	GA + log residual (trained pre-pandemic)	0.8881	1547	21360.0000
Fetal death	Post-pandemic (2022-2024)	GA + log residual (trained pre-pandemic)	0.8793	2315	30106.0000

Table 33: Growth-standard adjudication by out-of-sample mortality discrimination. Each standard's relative size-for-gestational-age deviation is entered (with its absolute value, plus GA and GA^2) into a common grouped-binomial model trained on early years and validated on later years; the reported weighted AUC measures how well each standard ranks perinatal mortality.

comparison	outcome	standard	validation weighted auc	n validation cells
External standards	Cohort infant death	Population (empirical, sex-specific)	0.7809	2000
External standards	Cohort infant death	WHO fetal growth (EFW)	0.7809	2000
External standards	Cohort infant death	INTERGROWTH-21st (EFW)	0.7814	2000

comparison	outcome	standard	validation weighted auc	n validation cells
External standards	Fetal death	Population (empirical, sex-specific)	0.8858	2041
External standards	Fetal death	WHO fetal growth (EFW)	0.8863	2041
External standards	Fetal death	INTERGROWTH-21st (EFW)	0.8849	2041
Population vs customized (NH White + NH Black cells)	Cohort infant death	Population (sex-specific)	0.7658	3029
Population vs customized (NH White + NH Black cells)	Cohort infant death	Customized (sex + maternal race)	0.7625	3029
Population vs customized (NH White + NH Black cells)	Fetal death	Population (sex-specific)	0.8761	3076
Population vs customized (NH White + NH Black cells)	Fetal death	Customized (sex + maternal race)	0.8762	3076

Table 34: Fraction of multiverse specifications supporting each headline conclusion, demonstrating stability of the growth-zone findings across analytic choices.

conclusion	fraction specs supporting	n specifications
C1: below-p10 RR ¿ 2 (both endpoints)	1.000	200
C2: fetal below-p10 RR ¿ infant below-p10 RR	1.000	200
C3: fetal above-p90 RR ¿ infant above-p90 RR	1.000	200
C4: fetal above-p90 RR ¿ 1	0.975	200

Table 35: Minimum, median, and maximum growth-zone rate ratios across all multiverse specifications.

metric	min	median	max	n
infant_below_p10_rr	2.513	3.138	3.691	200
infant_above_p90_rr	0.722	1.022	1.698	200
fetal_below_p10_rr	3.019	4.284	5.337	200
fetal_above_p90_rr	0.865	1.216	1.876	200

Table 36: BCA biomass reference (W_REF) reparameterization invariance: the expected-size curve fit (RMSE, MAPE) is unchanged across W_REF values, confirming W_REF is a scaling convention rather than a free determinant of fit.

sex	n w ref values	w ref range g	rmse g min	rmse g max	rmse g spread	mape percent spread
Female	5	3000-4000	13.7477	13.7477	0.0000	0.0000
Male	5	3000-4000	14.9629	14.9629	0.0000	0.0000

Table 37: Uploaded conceptual and literature PDF inventory.

Document	Pages
2021 Fetal growth curves is there a universal reference GRANTZ.pdf	18
A Method & Strategy For COVID-19 MICHAELSON.pdf	21
Fernando_Nishini_Weiss_Penultimate (6).pdf	21
Michaelson Huros Poster Oct 23 2024.pdf	1
On How Why When We Grow Old. HUANG SU CITTI MICHAELSON 2 13 2024.pdf	20
Research Ideas.pdf	21
SUPPLEMENT to The Binary Cellular Biology of Human Growth PAPERS I II and III MICHAELSON 1 17 2025.pdf	170
The Binary Cellular Biology of Human Growth I Quantifying Creation and Growth Body and Its Parts MICHAELSON 1 17 2025.pdf	27
The Binary Cellular Biology of Human Growth II Calculating Fetal Size MICHAELSON 1 17 2025.pdf	16
The Binary Cellular Biology of Human Growth III Normal and Abnormal Size & Growth MICHAELSON 1 17 2025.pdf	24

2 Supplementary Figures

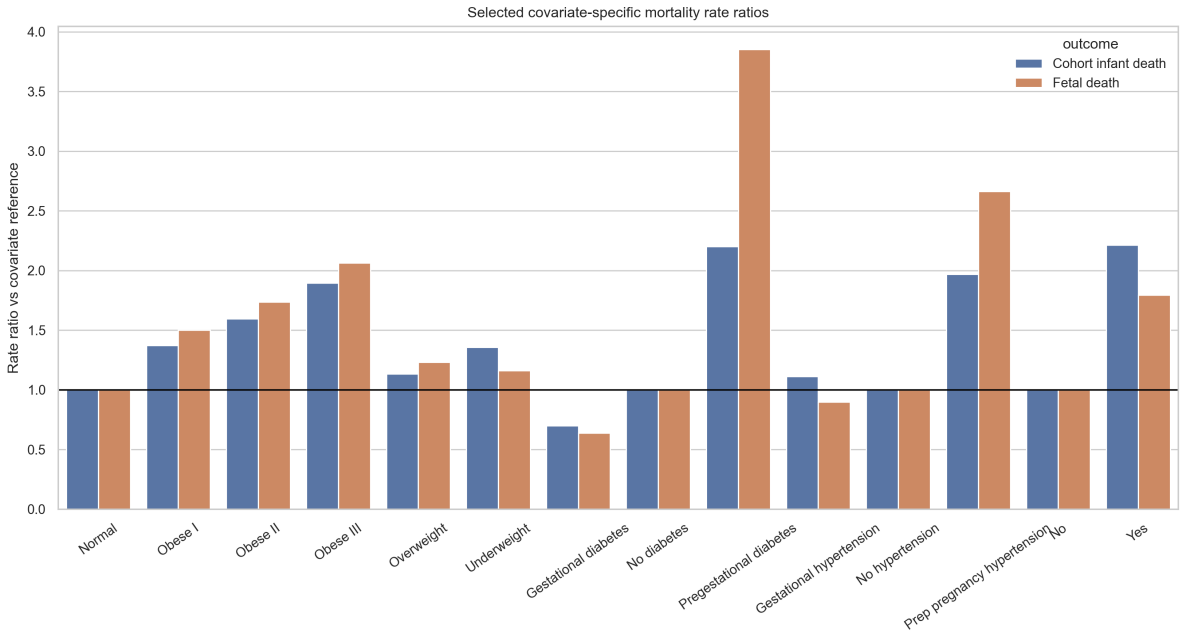


Figure 1: Selected descriptive covariate-specific mortality rate ratios. Race/ethnicity is treated as a social and clinical stratification variable, not a genetic explanation. These are not adjusted or causal estimates.

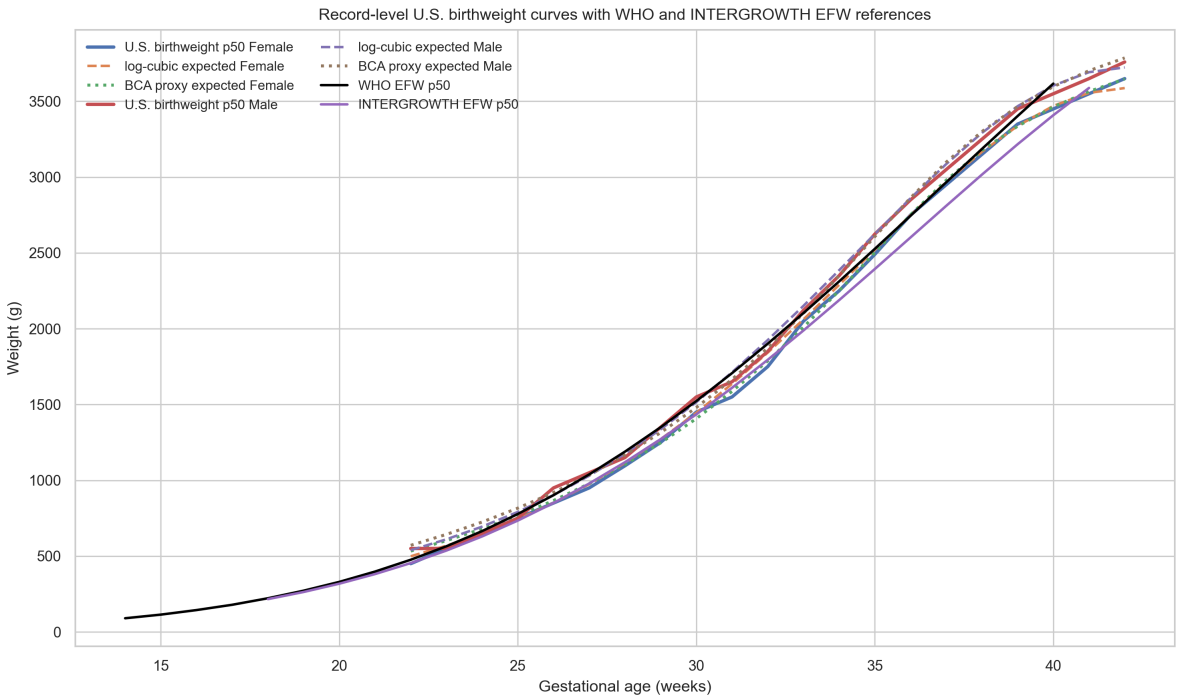


Figure 2: Record-level U.S. birthweight curves, log-cubic expected-size curves, BCA proxy expected-size curves, and WHO/INTERGROWTH estimated fetal-weight references. Birthweight at delivery and ultrasound EFW are not identical; this figure is a reference-alignment and model-comparison check, not a claim of interchangeability or a clinical ranking of standards.

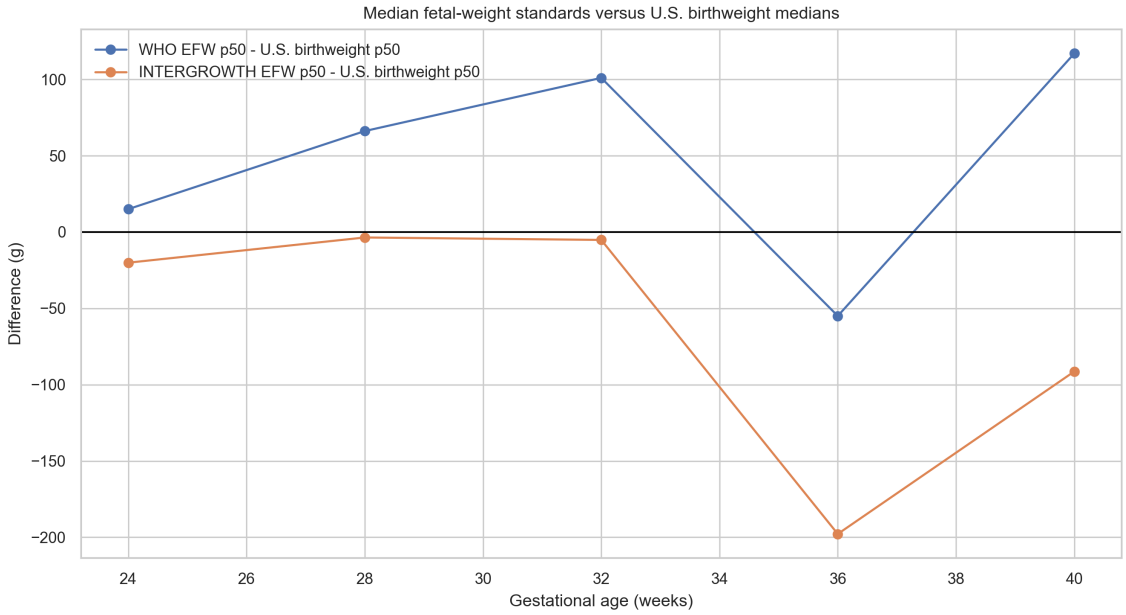


Figure 3: Differences between external fetal-weight reference medians and U.S. birthweight medians at selected gestational ages. This figure is a reference-alignment check, not a clinical ranking of standards.

3 Data Authenticity

The expanded analysis uses public source files and generated aggregate analysis tables. No simulated or invented outcome data are used.

Data authenticity audit:

`docs/provenance/expanded_data_authenticity_audit.md`

SHA-256 hashes:

`docs/provenance/expanded_analysis_hashes_sha256.txt`

4 Reproducibility

The expanded analysis is regenerated with:

```
python3 scripts/run_expanded_public_data_analysis.py
```

The stage-1 WONDER analysis is regenerated with:

```
python3 scripts/run_wonder_analysis.py
```

ZIP integrity is checked with:

```
python3 scripts/check_zip_integrity.py
```

References